Advanced Calculus

# Laplace transforms

## Understand

A Laplace transform represents a function of time using a new variable s. Linearity means that a sum can be transformed one term at a time. Constant multipliers stay outside the transform.

## Worked example

For s > 0, L{1} = 1/s and L{t} = 1/s^2. By linearity, L{2 + 3t} = 2/s + 3/s^2. Check each term before combining the result.

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# Try it, then check.

## Practice question

Find L{4 + 2t}, for s > 0.

## Answer and reasoning

4/s + 2/s^2. Apply linearity to the constant term and the term containing t.

## Reflect

Explain the steps in your own words. Create a review card and return to page 1 when you need the original explanation.